ModelIt!

Student Activity Pack

How Do We Build a Sustainable City?

Modeling Urban Systems, Energy Networks, and Resource Flows for Resilient Communities

Grade 12L3 | NGSS HS-ESS3-3 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What makes a city sustainable or unsustainable? List at least three specific factors and explain why each matters.**

2. **If you were redesigning your city to be more sustainable, what is the single change you would make first and why?**

3. **Why do cities consume so much more energy and produce so much more waste per area than rural regions?**

4. **What does equity have to do with sustainability — how can environmental improvements benefit some communities more than others?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Population Density
- Solar Radiation Availability
- Regional Water Supply
- Energy Mix Composition
- Transportation Mode Share
- Water Recycling Rate
- Waste Diversion Rate
- Green Infrastructure Coverage
- Carbon Emissions Rate
- Urban Sustainability Index

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Business as Usual

Energy: 30% renewable | Transport: 85% car | Waste Diversion: 35% | Green Space: 10%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Energy-Only Solution

Energy: 80% renewable | All other systems unchanged

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Integrated Redesign

Energy: 80% renewable | Transport: 60% transit/walk/bike | Water Recycling: 80% | Waste Diversion: 75% | Green Space: 30%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Food System Localization — The percentage of food consumed in the city that is grown within 100 miles — local food systems reduce transportation emissions, increase food security, create urban agricultural jobs, and connect residents to the ecological basis of their nutrition

Social Equity Index — A measure of how evenly sustainability benefits and environmental burdens are distributed across neighborhoods of different income levels and racial compositions — true sustainability requires that air quality, green space, transit access, and climate resilience do not correlate with wealth or demographic identity

Climate Resilience Score — The city's capacity to withstand and recover from climate-related disruptions including heat waves, flooding, drought, and extreme storms — resilience requires redundant infrastructure, natural buffers, emergency preparedness, and community social networks

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

By 2050, over 68% of humanity will live in cities. These urban centers consume 75% of global energy, produce 70% of carbon emissions, and generate billions of tons of waste annually. Can we redesign cities as self-sustaining ecosystems rather than resource-consuming machines, and what would that transformation require? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that renewable energy alone cannot make a city sustainable, and what additional system changes are required?**

2. **Why does transportation mode share have such a large impact on the Urban Sustainability Index when transportation is only one of many urban systems?**

3. **What does the model reveal about the relationship between green infrastructure coverage and the cost of engineered infrastructure solutions?**

4. **How would adding a Social Equity Index to your model change which sustainability interventions you prioritize and how you implement them?**

5. **What are the limitations of modeling an entire city as a ten-component system when real cities have millions of interacting variables and stakeholders?**

8 STEM Challenge: Design a Sustainable City Master Plan

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YOUR ENGINEERING MISSION

Design a comprehensive sustainability master plan for a city of 500,000 people that achieves net-zero carbon emissions by 2050 while improving livability, equity, and economic vitality, specifying quantitative targets for each urban system and justifying them with model evidence.

THE SCENARIO

The mayor of a mid-sized American city (population 500,000) has hired your urban planning team to create a sustainability master plan. The city currently has 25% renewable energy, 90% car mode share, 30% waste diversion, minimal green infrastructure, and aging water systems. The mandate: achieve net-zero carbon emissions by 2050 while ensuring that sustainability improvements benefit all residents equitably — not just wealthy neighborhoods. Budget constraints mean you cannot change everything at once. Your plan must prioritize, sequence, and justify every intervention.

Engineering Design Thinking

1. WHAT QUANTITATIVE TARGETS DOES YOUR PLAN SET FOR EACH URBAN SYSTEM BY 2030, 2040, AND 2050?

2. HOW DOES YOUR PLAN ENSURE THAT SUSTAINABILITY IMPROVEMENTS REACH LOW-INCOME NEIGHBORHOODS FIRST RATHER THAN LAST?

3. WHAT ECONOMIC INCENTIVES OR REGULATORY CHANGES DOES YOUR PLAN PROPOSE TO DRIVE THE TRANSITION?

4. HOW WOULD YOU MEASURE PROGRESS AND ADAPT THE PLAN IF TARGETS ARE NOT BEING MET?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

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Explain your design using scientific evidence from the simulation:

Real Career Connection

Urban planners who design sustainable communities earn $55,000-$110,000/year at city agencies and consulting firms. Environmental engineers who design water, waste, and energy systems for cities earn $70,000-$130,000/year. Sustainability directors who coordinate corporate and municipal sustainability programs earn $80,000-$150,000/year. Transportation planners who design transit systems and bike networks earn $60,000-$105,000/year.